

Adaptive Civic Routing Intelligence System

Deterministic PostGIS Jurisdiction Containment & Authority Routing

Paired with assistive Gemini AI for verified civic issue categorization.

TEAM NAME & SUBMISSION ID

Team Spark

HM26-E9C5

COLLEGE AFFILIATION

Acharya Institute of Tech.

Bengaluru, Karnataka

LIVE WORKING MVP & REPO

adaptive-civic-routing.vercel.app

React 18 + Node.js + PostGIS 3.6

Team Members: **Hemanth N** * **Karibasappa KL** * **Pooja M** * **Soumya Doddamani**

Branch: Master | Sub-problem: Routing | Repository: github.com/kkaribasappakl-eng/adaptive-civic-routing

Why Civic Complaints Get Stuck

Boundary ambiguity causes jurisdictional disputes between MCC and MUDA

REAL-WORLD MYSURU GROUND SCENARIO

Sewage burst on Bogadi Outer Ring Road sits unaddressed for 3 weeks.

How Jurisdiction Fails on Border Corridors:

- * MCC claims the road belongs to MUDA peripheral development.
- * MUDA claims maintenance was formally handed over to MCC Ward 42.
- * Result: Complaints bounce between departments with zero resolution.

Root Engineering Cause:

Current municipal systems rely on manual sorting or postal text matching. They lack deterministic spatial geometry to prove legal authority.

Mysuru Reality:

65 MCC wards with irregular, non-convex polygons cannot be solved with radial circles.

65 MCC Wards x MUDA Zones

- * 65 distinct MCC administrative wards.
- * Peripheral MUDA layouts in urban transition.
- * Frequent boundary disputes along Ring Road corridors.
- * No unified cross-agency GIS ledger currently exists.

Who Is Hurt Today?

- * Citizens: Trapped in closed tickets without escalation.
- * Ward Engineers: Waste time disputing jurisdiction.
- * Field Crews: Dispatched to incorrect ward locations.
- * City Leadership: Blind to cross-agency bottlenecks.
- * Public Trust: Eroded by delayed municipal accountability.

Built for Real Field Constraints

Context-specific interfaces for citizens, desk triage officers, and field workers

CITIZEN

Asha, 34

Resident in Kuvempunagar

DEVICE: Mobile Smartphone

NETWORK: Spotty 4G / 3G

LANGUAGE: Kannada & English

- * Needs to report in < 30 sec.
- * Cannot know ward boundaries.
- * Drops GPS pin on Leaflet map.
- * Uploads photo evidence.
- * AI suggests category instantly.
- * Receives tracking code on submit.

DESK OFFICER

Ramesh, 46

MCC Revenue & Triage Desk

DEVICE: Desktop Workstation

NETWORK: ~~Internet~~ Fiber 2G / 3G

LANGUAGE: ~~Kannada~~ Only Kannada

- * Needs mathematical proof.
- * Dispatches municipal budget.
- * Reviews unrouted complaints.
- * Inspects PostGIS ST_Covers.
- * Resolves border dispute cases.
- * One-click SLA escalation.

FIELD WORKER

Basavaraju, 38

Sanitary Inspector / Field Crew

DEVICE: Android Phone

- * Cannot use heavy desktop app.
- * Receives precise GPS pin.
- * Simple 1-tap navigation.
- * Clear Kannada task card.
- * Takes resolution photo.
- * Closes ticket with proof.

Core Routing Journey in 4 Steps

From citizen photo capture to verifiable municipal dispatch in under 50ms

01

Citizen Intake

- * Leaflet map pin placement
- * Photo evidence (5MB cap)
- * 100m proximity duplicate check

02

AI Classification

- * Gemini API assistant
- * 8-category vocabulary check
- * Citizen confirms or overrides

03

GIS Containment

- * PostGIS ST_Covers query
- * MultiPolygon ward matching
- * SRID 4326 spatial resolution

04

Audit & Dispatch

- * Code: HM-CIV-2026-XXXXXX
- * Assigned to MCC / MUDA
- * Locked to active version V1

LIVE WORKING MVP HERO RECEIPT (MATCHES VIDEO DEMO)

Automated PostGIS Dispatch Modal Generated on Submission:

- * Unique complaint code generated: HM-CIV-2026-XXXXXX
- * Deterministic containment: Assigned to MCC Ward 42
- * Department: Solid Waste Management (SLA: 24 Hours)
- * Zero AI in routing: 100% spatial PostGIS query execution



AI Assists. GIS Decides.

Strict architectural separation of concerns matching docs/architecture.md

ASSISTIVE AI LAYER (GEMINI API)

Issue Classification Assistance

- * Scope: Helps citizen pick civic category.
- * Analyzes description text + optional photo.
- * Controlled vocabulary check (server/src/services/aiService.js):

`GARBAGE, ILLEGAL_DUMPING, POTHOLE, DRAINAGE,
STREETLIGHT, C_AND_D_WASTE, WATER_LEAK, OTHER`

! STRICT ARCHITECTURAL PRINCIPLE:

AI NEVER decides jurisdiction or department.

Zero probabilistic routing hallucinations.

If API key unconfigured: manual category selection.

DETERMINISTIC GIS LAYER (POSTGIS)

Spatial Point-in-Polygon Engine

- * Coordinates stored in SRID 4326 geometry.
- * Wards modeled as true MultiPolygons.
- * Containment evaluated mathematically:

`ST_Covers(j.boundary, c.location)`

- * Boundary touch ambiguity detection:

`ST_Touches(j.boundary, c.location)`

* Immutable Jurisdiction Versioning:

Single active version (MYS_2026_V1).

Decisions permanently stored in routing_decisions.

DATA FLOW: Citizen Pin (Leaflet) --> Express Multer (5MB) --> PostGIS ST_Covers --> routing_decisions Table --> Socket.IO Real-Time Gateway

Hard Constraints Evaluation

Condensed docs/constraints.md: Honest evaluation of what works and known limits

CONSTRAINT	STATUS	OUR ARCHITECTURAL APPROACH	HONEST LIMITATIONS
1. Take / Fake Reports	✓ HANDLED	Leaflet GPS pin placement; 5MB MIME cap (JPEG, PNG, WebP); 100m proximity duplicate warning search without blocking.	Active on deployed MVP. Citizens can submit with or without photos.
2. Jurisdiction	✓ HANDLED	PostGIS ST_Covers point-in-polygon; ST_Touches border ambiguity check; immutable routing_decisions ledger on MYS_2026_V1	Zero AI in jurisdiction routing. 100% deterministic spatial math.
3. Priority & SLA	✓ HANDLED	Category severity matrix; state machine: SUBMITTED -> ROUTED -> IN_PROGRESS -> RESOLVED; Socket.IO live updates.	Database deadline calculation active. Operator escalation queue.
4. Bad Input	✓ HANDLED	Strict Mysuru bounding box check (Lat 12.2-12.4, Lng 76.5-76.8); out-of-bounds coordinates quarantined to HUMAN_REVIEW	Invalid coordinates and script uploads rejected at API layer.
5. Offline Operation	! PARTIAL	Graceful degradation: system functions without AI if key missing; clear offline UI banners; 503 database health checks.	Requires connection to submit. Local Service Worker sync in Phase 2.

Key Engineering Decisions

Condensed ADR Q1 + Q2: Evaluating trade-offs, rejected alternatives, and costs accepted

DECISION Q1: POSTGIS VS PROBABILISTIC LLM

CHOSEN APPROACH:

PostgreSQL 18 + PostGIS 3.6 spatial geometry engine.
Evaluates ST_Covers on MultiPolygon ward boundaries.

REJECTED ALTERNATIVES:

- * LLM text classification (hallucinates on border zones).
- * Radial circle distances (Mysuru wards are non-convex).
- * Postal PIN codes (too coarse; 1 PIN spans multiple wards).

TRADE-OFF & COST ACCEPTED:

- * Must ingest and maintain GIS shapefiles from MCC/MUDA.
- * Requires GiST spatial indexing overhead in PostgreSQL.

=> **GAIN: 100% mathematically reproducible and legally defensible.**

DECISION Q2: IMMUTABLE VERSIONING VS IN-PLACE

CHOSEN APPROACH:

Decoupled lifecycle: DRAFT -> ACTIVE -> RETIRED.
Tickets permanently locked to active version MYS_2026_V1.

REJECTED ALTERNATIVES:

- * In-place polygon overwrites (destroys historical audit).
- * Re-routing closed historical tickets when wards change.
- * Multiple concurrent active versions (causes conflicts).

TRADE-OFF & COST ACCEPTED:

- * Additional table storage for historical boundary snapshots.
- * Schema enforces uq_single_active_version constraint.

=> **GAIN: Municipal delimitation changes preserve legal proof.**

Proof It Works — Real MVP Evidence

Genuine captures from deployed production instance (adaptive-civic-routing.vercel.app)

1. Normal Intake Flow

Photo upload & AI category tag



What this proves:

- * Genuine Leaflet map GPS pin.
- * Real Gemini AI category tag.
- * Proximity duplicate badge.
- * Non-blocking intake UX.

2. PostGIS Routing Result

Instant municipal dispatch modal



What this proves:

- * Tracking code HM-CIV-2026.
- * Contained in MCC Ward 42.
- * Dispatched to SWM Dept.
- * Executed in sub-50ms.

3. Flagged Border Case

Human Review Quarantine

```
table: routing_decisions
routing_status: 'HUMAN_REVIEW'
review_status: 'OPEN'
active_version: 'MYS_2026_V1'
audit: complaint_review_actions
```

What this proves:

- * Out-of-bounds halted at API.
- * Quarantined for operator review.
- * Append-only action audit trail.
- * Zero silent routing failures.

Engineering Realism & Scale Limits

Top 3 things that break at city scale and the architectural fixes (Decision Log Q3)

BREAKPOINT 01

Monsoon Surge Load

What Breaks:

- * 10,000 concurrent reports during Mysuru flood events.
- * Repeated ST_Covers queries saturate database CPU.

Architectural Fix:

1. Bounding-box pre-filtering (&&) before polygon calculation.
2. BullMQ asynchronous queue buffers intake during spikes.
3. Read replica for public lookups.

BREAKPOINT 02

Historic Ticket Scale

What Breaks:

- * Over 5 years, complaints exceed 10 million rows.
- * GiST spatial index size outgrows server RAM.

Architectural Fix:

1. Declarative range partitioning by fiscal year and ward.
2. Cold-storage archival for resolved historical tickets.
3. Keeps active index in RAM.

BREAKPOINT 03

Multi-Agency Overlaps

What Breaks:

- * Shared border roads between MCC and MUDA layouts.
- * Ambiguous boundaries on median or outer ring road.

Architectural Fix:

1. ST_Touches boundary detector.
2. Quarantines to Human Review with split SLA attribution.
3. Inter-agency MOU protocol.
4. Joint inspection dispatch.

What Comes Next & Disclosure

Pilots, civic data partnerships, and transparent AI tool disclosure

3 CONCRETE NEXT STEPS

Step 1: Ward-Level Pilot (MCC Ward 42)

Field deployment with MCC Ward 42 corporator desk and sanitary inspectors in Bogadi and Kuvempunagar.

Step 2: Official Mysuru GIS Survey Ingestion

Partner with Mysore Urban Development Authority (MUDA) to ingest certified master plan shapefiles and parcel boundaries.

Step 3: Kannada Voice & Offline Sync

Local Service Worker store-and-forward queue with Kannada speech-to-text for field workers with low connectivity.

AI TOOLS DISCLOSURE

- * Tool Used: Google Gemini API.
- * Role: Civic issue category suggestion only.
- * Zero AI used for jurisdiction or routing.
- * Full details documented in docs/architecture.md.

VERIFIED SUBMISSION LINKS

PRODUCTION WORKING MVP:

<https://adaptive-civic-routing.vercel.app>

GITHUB REPOSITORY:

github.com/kkaribasappakl-eng/adaptive-civic-routing

10-MINUTE WALKTHROUGH VIDEO:

drive.google.com/file/d/1Jq3ZSU160H3UGLkgwKjcmzYTMe3CY3Kx